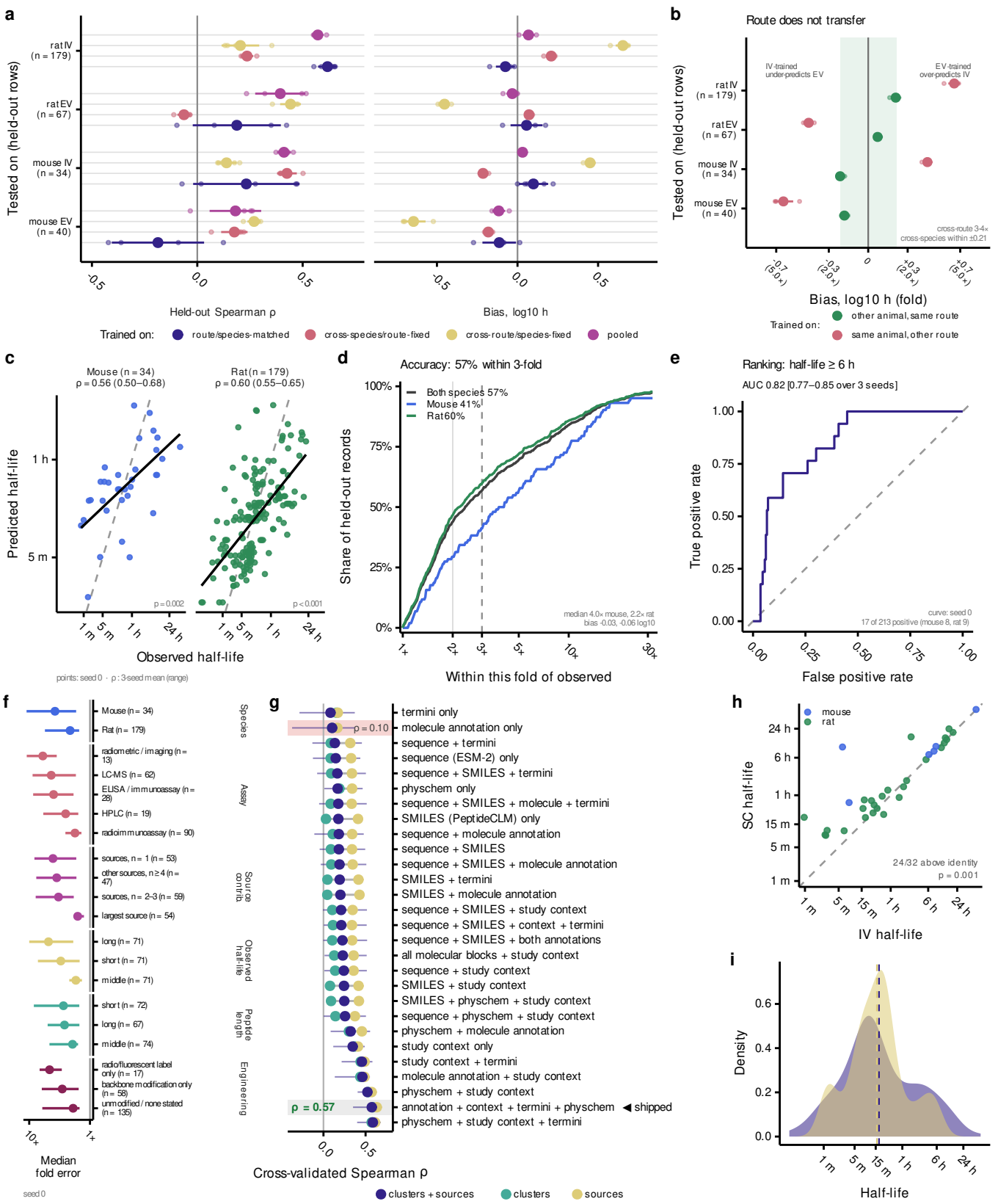

a
Tested on (held-out rows)
rat IV (n = 179)
rat EV (n = 67)
mouse IV (n = 34)
mouse EV (n = 40)
Held-out Spearman ρ
Bias, log10 h
Trained on:
route/species-matched
cross-species/route-fixed
cross-route/species-fixed
pooled
b
Route does not transfer
IV-trained under-predicts EV
EV-trained over-predicts IV
cross-route 3-4×
cross-species within ±0.21
Bias, log10 h (fold)
Trained on:
other animal, same route
same animal, other route
c
Mouse (n = 34)
ρ = 0.56 (0.50–0.68)
Rat (n = 179)
ρ = 0.60 (0.55–0.65)
Predicted half-life
Observed half-life
p = 0.002
p < 0.001
points: seed 0 · ρ: 3-seed mean (range)
d
Accuracy: 57% within 3-fold
Both species 57%
Mouse 41%
Rat 60%
Share of held-out records
median 4.0× mouse, 2.2× rat
bias -0.03, -0.06 log10
Within this fold of observed
e
Ranking: half-life ≥ 6 h
AUC 0.82 [0.77–0.85 over 3 seeds]
True positive rate
False positive rate
curve: seed 0
17 of 213 positive (mouse 8, rat 9)
f
Species
Mouse (n = 34)
Rat (n = 179)
Assay
radiometric / imaging (n = 13)
LC-MS (n = 62)
ELISA / immunoassay (n = 28)
HPLC (n = 19)
radioimmunoassay (n = 90)
Source contrib.
sources, n = 1 (n = 53)
other sources, n ≥ 4 (n = 47)
sources, n = 2–3 (n = 59)
largest source (n = 54)
Observed half-life
long (n = 71)
short (n = 71)
middle (n = 71)
Peptide length
short (n = 72)
long (n = 67)
middle (n = 74)
Engineering
radio/fluorescent label only (n = 17)
backbone modification only (n = 58)
unmodified / none stated (n = 135)
Median fold error
seed 0
g
termini only
ρ = 0.10
molecule annotation only
sequence + termini
sequence (ESM-2) only
sequence + SMILES + termini
physchem only
sequence + SMILES + molecule + termini
SMILES (PeptideCLM) only
sequence + molecule annotation
sequence + SMILES
sequence + SMILES + molecule annotation
SMILES + termini
SMILES + molecule annotation
sequence + SMILES + study context
sequence + SMILES + context + termini
sequence + SMILES + both annotations
all molecular blocks + study context
sequence + study context
SMILES + study context
SMILES + physchem + study context
sequence + physchem + study context
physchem + molecule annotation
study context only
study context + termini
molecule annotation + study context
physchem + study context
ρ = 0.57
annotation + context + termini + physchem
◀ shipped
physchem + study context + termini
Cross-validated Spearman ρ
clusters + sources
clusters
sources
h
mouse
rat
SC half-life
IV half-life
24/32 above identity
p = 0.001
i
Density
Half-life